

RESEARCH INTERESTS

Machine Learning, Natural Language Processing, Multilingual and Low-Resource NLP, AI Safety, Multimodal Learning, Clinical Decision Support Systems

EDUCATION

Brown University Ph.D. in Computational Biology, Advisor: Hamish Fraser M.S. in Computer Science, GPA: 3.9/4.0	Providence, USA 2023–Present 2021–2023
Nelson Mandela -AIST M.S. in Information and Communication Science and Engineering, GPA: 4.34/5.00	Arusha, Tanzania 2012–2014
Saint Augustine University of Tanzania B.S. in Computer Science, GPA: 4.50/5.00	Iringa, Tanzania 2006–2009

EXPERIENCE

Graduate Research Assistant Brown University	Providence, USA 2023–Present
- Conduct research on the evaluation, robustness, and safety of machine learning, deep learning, and large language model (LLM)–based clinical decision support systems. - Develop multilingual and low-resource evaluation benchmarks to assess reliability, hallucination risk, and policy compliance in healthcare settings.	
Data Scientist Center for Financial Inclusion (Cenfri)	Remote, South Africa March–November 2018
Led the development of a machine learning–based credit scoring system to improve financial access for underserved rural communities in Tanzania.	
Assistant Lecturer College of Business Education	Dodoma, Tanzania 2010–2021
- Taught undergraduate courses in programming (C/C++), business mathematics, and supervised capstone research projects. - Mentored students in applied machine learning and software development projects.	

PUBLICATIONS

- [1] T. Abdullahi, **M. Mgonzo**, M. Oduwole, P. Okewunmi, A. Owodunni, R. Singh, and C. Eickhoff, “Ubuntuguard: A Culturally-Grounded Policy Benchmark for Equitable AI Safety in African Languages”, *ACL*, 2026.
- [2] **M. Mgonzo**, K. Oketch, N. A. Etori, W. Mang’eni, E. F. Nyaki, and M. S. Mollel, “Learning from Scarcity: Building and Benchmarking Speech Technology for Sukuma.”, in *Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026)*, ACL, 2026.

[3] R. Esfandiarpour, G. Zerveas, R. Zhang, **M. Mgonzo**, C. Eickhoff, and S. Bach, “Beyond Contrastive Learning: Synthetic Data Enables List-wise Training with Multiple Levels of Relevance”, in *EMNLP*, 2025.

PROFESSIONAL SERVICE

- **Reviewing**
 - Machine Learning for Healthcare (ML4H) Conference 2023–2026
 - COLM Workshop on Deploying AI in healthcare (DAIH) 2026
 - Black in AI (BAI) Workshop 2024
 - Deep Learning Indaba (DLI) Conference 2024, 2025
- **Volunteering**
 - ACL 2026
 - SIGIR 2022
 - FAccT 2022
- **Organizing**
 - IndabaX, Nelson Mandela AIST, Arusha, Tanzania 2020

TEACHING

- **Graduate Teaching Assistant** at Brown University Fall 2024
 - Evaluation of Health Information Systems (BIOL 2075)*
- **Assistant Lecturer** at College of Business Education 2010–2021
 - Advanced Business Mathematics, Programming in C, C++ (SMU 7304,ITT 5102 & ITT 6102)*

SKILLS

- **Programming:** Python (advanced), C/C++, SQL.
- **ML & AI:** PyTorch, TensorFlow, HuggingFace, Scikit-learn.
- **Developer Tools:** Git, Linux, LaTeX.
- **Big Data Analytics Frameworks:** Hadoop, MapReduce.
- **Languages:** English Fluent; Swahili Native Speaker.

SELECTED PROJECTS

- **Cross-Modal Alignment for Unpaired Multimodal Data** 2025
 - Tools used: ViTs, BERT, Gromov-Wasserstein Optimal Transport*
- **Systematicity in Recurrent Transformers** 2023
 - Tools used: Universal Transformer*

SCHOLARSHIPS AND AWARDS

- Black in AI Travel Award (NeurIPS 2022) 2022
- Google CS Research Mentorship Program (CS-RMP) 2021
- ACM Travel Grant (FAccT 2019, FAccT 2020) 2019, 2020
- Google Conference Travel Award(DLI 2018) 2018